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"Learning is repeated cyclic process"

Machine Learning
⇓

!! let the Machine learn from
"DATA" !!

But What Machine Trying to
learn from data :- Pattern / Trend / etc

How Machine Learning Learns from Data

Suppose : you need to predict house price



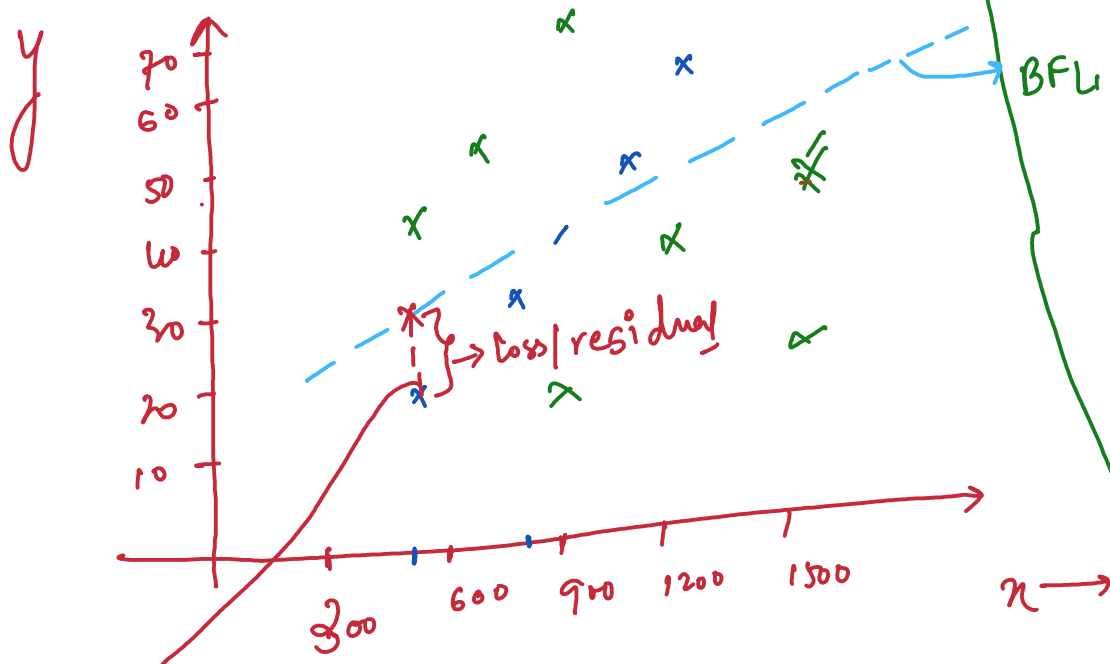
sample
dataset

Size (sq. ft.)	price (INR) lakh
500	20
800	32
1000	40
1200	48
1500	60
...	...

$x \rightarrow$ Input $\Rightarrow$ Independent
predictor

$y \rightarrow$ output $\Rightarrow$ dependent
feature

Problem
Re
An
wh
Bas



$$\text{loss} \Rightarrow (\hat{y} - y)$$

$$\text{loss} = \sum \text{loss}$$

Line with minimum error/loss

Regression Eq.

hypothesis

n ... n ... n ...

$$h_{\theta}(x) = \sum_{i=0}^n \theta_i x_i, \quad x_0 = 1$$

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots$$

model / hypothesis / predicted value.

we also rep. this $\Rightarrow \hat{y}$

size x	price y
5	20
8	32
10	40
12	48
15	60

Randomly we initialize θ 's

$$\theta_0 = 0$$

$$\theta_1 = 0.5$$

$$\hat{y} = h_{\theta}(x) = \theta_0 + \theta_1 x_1 = 0 + 0.5 \times 5 = 2.5$$

Multiple loss fn
MSE

$$\text{Loss} \Rightarrow \frac{1}{2 \times 1} (2.5 - 20)^2 = \frac{1}{2} \times (-17.5)^2 = 153.125$$

As a client will you take this m with large loss. ??

No, then how to improve this mod
 Gradient Descent Comes

$$J(\theta) = \frac{1}{2m} \sum_{i=0}^m (\hat{y}^{(i)} - y^{(i)})^2 \quad \text{Cost fn} \Rightarrow \text{Sum}$$

or

$$J(\theta) = \frac{1}{2} (\hat{\theta}_1 x - y)^2$$

We have take $\hat{\theta}_1$ give = constant

minimize $\Rightarrow$ diffent

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{2} \times 2 (\theta_1 x - y) \cdot x$$

Gradient

$$= (0.5 \times 5 - 20) 5$$

$$= -87.5$$

⊗ Update $\theta_1 \Rightarrow$ Learnable parameters. 0.1 - 0.01

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \frac{\partial J}{\partial \theta_{\text{old}}}$$

$$= \underline{0.5} - \underline{0.01} \times (-87.5)$$

$$= 0.5 + 0.875$$

$$= \boxed{1.375}$$

$$\begin{array}{l} * \quad \theta_{old} = 0.5 \\ \theta_{new} = 1.375 \end{array}$$

⇒ This is called learning

Let's now the op with new θ .

$$\Rightarrow \hat{y} = h_0(x) = \theta_0 + \theta_1 x \Rightarrow 0 + 1.375x = 6.875$$

$$\text{Loss} = \frac{1}{2 \times 1} (6.875 - 20)^2$$

$$= \frac{1}{2} \times (-13.125)^2$$

$$= \boxed{87.133}$$

* Old loss $\Rightarrow$

$$\boxed{153.125}$$

New loss $\Rightarrow$

$$\boxed{87.133}$$

Improvement

This process will repeat till θ will v

$$\theta_{\text{Old}} \approx \theta_{\text{new}} \quad \swarrow \quad | \quad \text{stagnate}$$

Convergence $\Rightarrow$ Best

$\theta \Rightarrow$ final Value $\Rightarrow$ Think

Initially $\Rightarrow$ we started with random θ

means $\left\{ h_{\theta}^{(2)} = \theta_0 + \theta_1 x \right\}$
 one line θ_1 $\leftarrow$ next iteration and θ will con

new line will come $\leftarrow$

Till.
 your model will not get best fit

Gradient Descent $\Rightarrow$ Model Optimization Technique

MSE Evaluation Metric for Regression

Model Eval = = =

1) MSE $\Rightarrow$ Mean Squared Error

$$\Rightarrow \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i)^2$$

② MAE $\Rightarrow$ Mean Absolute Error

$$\frac{1}{2m} \sum_{i=1}^m |h_{\theta}(x_i) - y_i|$$

③ RMSE $\Rightarrow$ Root Mean Square Error

$$= \sqrt{\frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i)^2}$$

$$\textcircled{4} \quad R^2\text{-score} = 1 - \frac{\sum (h_{\theta}(x_i) - y_i)^2}{\sum (\bar{y} - y_i)^2}$$

$\downarrow$
Closer to 1 $\Rightarrow$ Best Model

Attribute Information:

- instant: record index
- dteday : date
- season : season (1:winter, 2:spring, 3:summer, 4:fall)
- yr : year (0: 2011, 1:2012)
- mnth : month (1 to 12)

- hr : hour (0 to 23)
- holiday : weather day is holiday or not (extracted from [Web Link])
- weekday : day of the week
- workingday : if day is neither weekend nor holiday is 1, otherwise is 0.
- weathersit :
- 1: Clear, Few clouds, Partly cloudy, Partly cloudy
- 2: Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist
- 3: Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds
- 4: Heavy Rain + Ice Pallets + Thunderstorm + Mist, Snow + Fog
- temp : Normalized temperature in Celsius. The values are derived via $(t-t_{\min})/(t_{\max}-t_{\min})$, $t_{\min}=-8$, $t_{\max}=+39$ (only in hourly scale)
- atemp: Normalized feeling temperature in Celsius. The values are derived via $(t-t_{\min})/(t_{\max}-t_{\min})$, $t_{\min}=-16$, $t_{\max}=+50$ (only in hourly scale)
- hum: Normalized humidity. The values are divided to 100 (max)
- windspeed: Normalized wind speed. The values are divided to 67 (max)
- casual: count of casual users
- registered: count of registered users
- cnt: count of total rental bikes including both casual and registered

